DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) | PHASE 5

Customer Churn Prediction & Retention Analytics

Project Planning & Management — Part 1 Documentation

PROGRAM	MODULE	TARGET DATE	REPOSITORY
DEPI Phase 5 Data Science Track	Part 1 – Project Planning & Management	October 16, 2026	GitHub (documents in /docs)

1 Project at a Glance

An end-to-end analytics and machine-learning solution for a subscription-based company that is losing customers without knowing who is likely to leave. The team will query customer data with SQL, clean and explore it, train and compare four binary classification models, and deliver an interactive dashboard with a high-risk customer list that supports targeted retention campaigns.

4	4	5	≥ 85%
Team Members	Classification Models Compared	Dashboard Views	Target F1-Score

2 Team Structure & Task Assignment

Member Name	Role / Specialization	Primary Responsibilities
Ebrahim Hassan TEAM LEADER	**Dashboard Developer & Team Leader**	Build the Plotly/Dash dashboard (churn rate, churn by segment and subscription type, trends, high-risk customers). Supervise the project: repository management, pull-request reviews, and progress tracking.
Ahmed Hameed	**ML Engineer & Modeling Lead**	Train and compare Logistic Regression, KNN, Decision Tree, and SVM using Scikit-Learn, handle class imbalance with SMOTE, and select the best model using Accuracy, Precision, Recall, and F1-Score.
Rahma Yasser	**Data Analyst & EDA Lead**	Clean the data (missing, inconsistent, and outlier values) and perform EDA with Pandas and Seaborn to find the factors linked to churn.
Habiba Zalam	**Data Engineer & SQL Lead**	Set up the relational database (Customers, Subscriptions, Transactions, Support) and write SQL queries that produce one unified customer table.

3 Resource Allocation

Resource Area	Tools & Technologies	Primary Owner
Data Layer	Relational database + SQL (customer, subscription, transaction, support data)	Habiba
Analysis	Python, Pandas, Jupyter Notebook (cleaning, EDA, feature engineering)	Rahma
Machine Learning	scikit-learn: Logistic Regression, KNN, Decision Tree, SVM; SMOTE for balancing	Ahmed
Dashboard	Plotly / Dash interactive business dashboard	Ebrahim

Resource Area	Tools & Technologies	Primary Owner
Collaboration	GitHub repository, GitHub Projects (Kanban), pull requests	Leader + all members

4 Project Proposal: Problem Statement

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cost of acquiring vs. retaining

A subscription-based company is losing customers and does not know which customers are most likely to leave or which characteristics are associated with churn. Every lost customer is lost recurring revenue, and retaining an existing customer costs roughly five times less than acquiring a new one.

The need: an automated, data-driven way to detect churn patterns early and predict at-risk customers, so retention budgets are spent on the right people at the right time.

5 SMART Objectives

S	**Specific**	Build an end-to-end binary classification solution (churn vs. retained) combining SQL analysis, data cleaning, EDA, model comparison, an interactive dashboard, and customer-level churn predictions.
M	**Measurable**	Reach an F1-Score ≥ 85% on the held-out test set and support a 15% churn-reduction target through targeted retention actions.
A	**Achievable**	Compare Logistic Regression, KNN, Decision Tree, and SVM; handle class imbalance with SMOTE. Ensemble models (Random Forest, XGBoost) are an optional stretch.
R	**Relevant**	Produce a high-risk customer list and segment insights that feed targeted retention marketing campaigns.
T	**Time-bound**	All deliverables follow the DEPI schedule, ending with the final presentation, testing, and reports on December 4, 2026.

6 Scope Boundaries

✔ IN SCOPE

- Relational customer database and SQL analysis
- Data cleaning and Exploratory Data Analysis (EDA)
- SMOTE class balancing
- Training and comparing four classification models
- Interactive Plotly/Dash dashboard with high-risk customer list
- Data-driven retention strategies and final business report

✘ OUT OF SCOPE

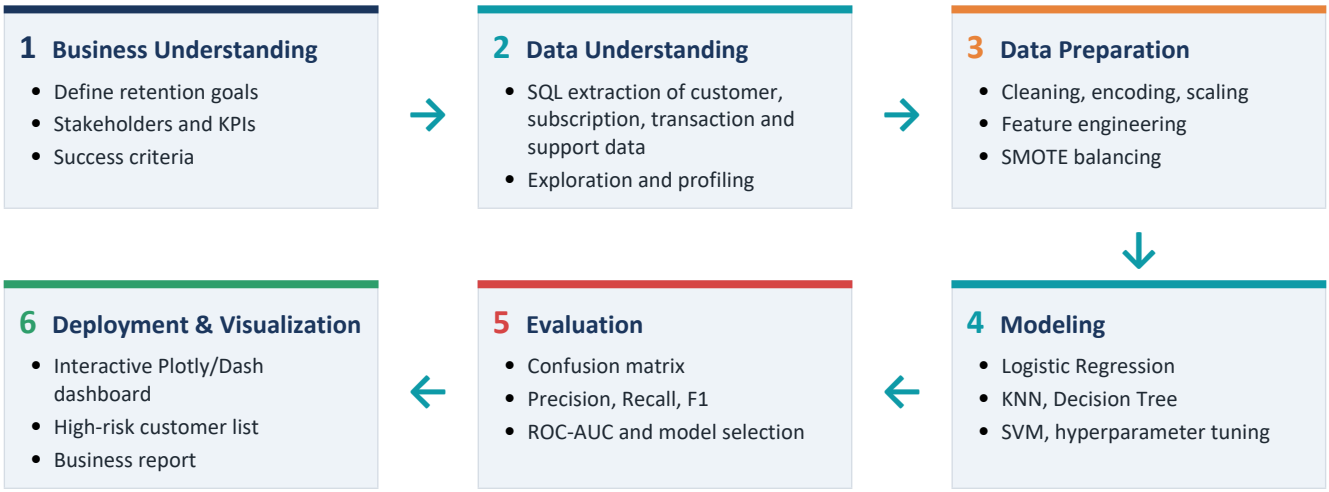
- Live enterprise CRM database integration
- Real-time automated SMS / Email marketing dispatch
- Production-scale deployment (MLflow + FastAPI are an optional extension only if time permits)

7 Key Performance Indicators (KPIs)

KPI	Target	How It Is Measured
Model F1-Score	**≥ 85%**	Best model on the held-out test set, compared across all four algorithms
Churn Recall	**≥ 80%**	Share of actual churners correctly identified (confusion matrix)
Churn Reduction	**15%**	Business target supported by the high-risk list and retention campaigns
Dashboard Coverage	**5 / 5 views**	Churn rate, by segment, by subscription type, trends, high-risk list
On-Time Delivery	**6 / 6**	Milestones completed by the dates in the project plan
Code Quality	**100%**	Merges into main made through reviewed pull requests

Targets are team-proposed and will be validated once the dataset has been profiled.

8 Methodology: CRISP-DM Lifecycle



Iterative cycle: evaluation results feed back into data preparation and business understanding until the F1 target is met.

9 Risk Assessment & Mitigation Matrix

Risk Category	Identified Risk	Impact	Mitigation Strategy
Technical	Severe class imbalance (few churners)	**High**	Apply SMOTE oversampling on training data only, plus algorithmic class weights; evaluate with F1, Recall and ROC-AUC rather than accuracy.
Model Performance	F1 target of 85% not reached	**High**	Feature engineering, hyperparameter tuning, and an optional ensemble (Random Forest / XGBoost).
Version Control	Merge conflicts and overwrites	**Medium**	Strict feature branching (feature/*) and mandatory pull-request approvals.
Schedule	Task delays	**Medium**	Weekly GitHub Projects tracking and task reallocation by the Team Leader.
Data Quality	Missing values and outliers	**Low**	Automated imputation pipelines and robust scaling.

10 Project Plan: Gantt Chart

Task (week starting)	Oct 12	Oct 19	Oct 26	Nov 2	Nov 9	Nov 16	Nov 23	Nov 30
Planning, proposal & team registration								
Literature review & requirements								
System analysis & design								
Data collection, SQL & preprocessing								
Exploratory Data Analysis (EDA)								
Model training & evaluation								
Dashboard & infographic								
Testing, QA & final report								
Final presentation & defense								

DEPI deadlines: Literature Review & Requirements Oct 16 | System Analysis & Design Nov 6 | Implementation Nov 30 | Final Presentation, Testing & Reports Dec 4

11 Milestones & Deliverables

#	Milestone	Date	Owner
M1	Documentation Part 1 submission (PDF in docs/)	Oct 16, 2026	Ebrahim
M2	Data preprocessing & EDA complete	Nov 13, 2026*	Rahma & Habiba
M3	Model training & evaluation complete	Nov 20, 2026*	Ahmed
M4	Interactive dashboard & infographic	Nov 27, 2026*	Ebrahim
M5	Final delivery & oral defense	Dec 04, 2026	All members

** Proposed team dates, to be confirmed. Other dates follow the DEPI schedule.*

12 Git / GitHub Collaboration Workflow



✘ Direct commits to main are prohibited	✔ Feature branches are mandatory for every task
✔ Descriptive commit messages are required	✔ Pull requests are reviewed by the Team members before merging
✔ All tasks are tracked on the GitHub Projects Kanban board	✔ All documents are uploaded to GitHub (docs/ folder)